Sentiment Analysis in Tennis: Correlating Sentiment Scores with Match Outcomes and Odds Movements

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Abstract

This study examines the correlation between social media sentiment and professional tennis match outcomes. The research employs data mining techniques to analyse sentiment dynamics by utilising Twitter data (3,515 tweets) and a combined dataset of ATP/WTA match results and betting odds for 180 matches from the 2025 season. Sentiment scores were derived using VADER, with tweets processed via Python libraries such as pandas and vaderSentiment. K-means clustering, implemented through scikit-learn, identified patterns in sentiment differences and odds, while Seaborn and Matplotlib facilitated visualisation. Findings reveal distinct sentiment patterns: ATP matches exhibit stronger reactions, with High Sentiment for Underdogs Won, whereas WTA matches show higher Neutral/Zero sentiment. Clustering distinguished high and low volatility matches, predicting upsets, particularly in ATP. Profit return analysis highlighted lucrative betting opportunities, such as Low Sentiment bets on ATP's Won Set 3 and High Sentiment bets on WTA's Won Set 3. These insights offer significant implications for sports betting, enabling traders to leverage fan sentiment to anticipate odds fluctuations and identify undervalued players, thus enhancing profitability in tennis betting markets.

Contents

1. Introduction

Sentiment analysis, a powerful tool in data mining, has transformed how we interpret public opinion across domains, yet its application in sports remains underexplored. In financial markets, sentiment indicators derived from social media often correlate with price movements, reflecting collective investor emotions (Bollen et al., 2023). This parallel suggests that public sentiment, as expressed on platforms like Twitter, could similarly influence or reflect outcomes in sports, where fan and bettor emotions drive engagement and market dynamics. Professional tennis, with its high-stakes ATP and WTA tournaments, offers a rich context to investigate whether sentiment scores correlate with match results, akin to sentiment-driven inefficiencies in betting markets (Forrest and Simmons, 2024). Understanding such correlations could reveal patterns that inform sports trading strategies and enhance analytical models.

This study aims to apply data mining techniques to examine correlations between Twitter-derived sentiment scores and match outcomes in professional tennis. By analysing sentiment data for players across match and set outcomes, including winners, losers, favourites, and underdogs, the study aims to identify whether positive or negative sentiment aligns with performance, similar to how market sentiment predicts financial trends. The significance of this research lies in its potential to provide actionable insights for sports traders, analysts, and enthusiasts. For traders, sentiment correlations could highlight betting market inefficiencies, whilst analysts and fans gain a deeper understanding of how public perception reflects or influences player performance (Mahmoud, 2024).

2. Data Collection & Understanding

2.1 Data Collection

Two integrated datasets are utilised in this study to explore correlations between Twitter sentiment and professional tennis match outcomes: Twitter social media data and a combined dataset of ATP/WTA match results and betting odds. The analysis encompasses 180 professional tennis matches (80 ATP, 100 WTA) from the Miami 2025 tournament season, selected for their prominence and active Twitter engagement, to investigate sentiment dynamics in high-profile tournaments.

2.1.1 Twitter Data

Twitter data was collected using Tweepy, interfacing with Twitter's Basic V2 API, which imposes a 15,000 tweets/month limit and restricts access to the search_recent_tweets endpoint (7-day historical data) following the discontinuation of the free API in 2023 (Twitter, 2023). A total of 3,515 tweets were scraped for 180 matches (approximately 1,563 for ATP, 1,952 for WTA), using the query "{player}" -is:retweet to prioritise original content. Tweets were collected during the duration of each set played in the matches, ensuring alignment with in-game events. Features include tweet text, timestamp, and VADER-derived compound sentiment scores ranging from -1.0 to +1.0, which will be categorised as High (≥ +0.05), Low (≤ -0.05), or Neutral/Zero (-0.05 < x < 0.05 or 0.0) for analysis. The tweet volume averages ~19.5 tweets per match (3,515 ÷ 180).

2.1.2 Tennis Match and Betting Odds Data

Tennis match and betting odds data were sourced from a private structured database, stored within the same MongoDB documents for 180 matches from the 2025 season. Each document includes:

- **Match Features:** Player names, match winner, date, tournament, set outcomes (winner of each set), and total sets (2 or 3).
- **Odds Features:** Starting odds (pre-match odds for each player), end-of-set odds (at the conclusion of Sets 1 and 2), start-of-set odds (at the start of Set 2, with a fallback to end-of-set odds if missing), and timestamps.

Odds are in decimal format (validated within 0.01-1000), aggregated from 500-3,000 records per player per match. This combined dataset provides a comprehensive record of match outcomes and betting market dynamics, enabling detailed analysis of sentiment's influence on performance and betting trends, akin to financial market inefficiencies (Forrest and Simmons, 2024).

2.2 Justification

Twitter data captures real-time fan sentiment during match sets, ideal for dynamic analysis of in-game sentiment, whilst the combined ATP/WTA match and odds dataset provides definitive outcomes and market expectations. The integration of 3,515 tweets and 180 match documents with embedded odds records enables robust correlation analysis between sentiment, performance, and betting trends. The API's tweet and 7-day time constraints necessitated focusing on high-profile 2025 tournaments with sufficient tweet volume, ensuring data quality within the Basic V2 API limits.

2.3 Ethical Considerations

Tweet data was anonymised, excluding user identifiers, to comply with GDPR and Twitter API terms (European Union, 2016; Twitter, 2023). Ethical scraping adhered to rate limits (450 requests/15-min) and terms of service. Sentiment score reliability was verified through manual tweet sampling, confirming VADER's applicability to sports contexts (Hutto and Gilbert, 2014). Match and odds data from the private database were used with permission, ensuring compliance with data privacy standards.

2.4 Dataset Summary

Dataset	Size	Features	Data Types

Twitter	3,515 tweets (~1,563 ATP, ~1,952 WTA)	Text, timestamp, sentiment score	String, datetime, float
Match & Odds	180 matches (80 ATP, 100 WTA)	Player names, winner, date, tournament, sets, starting odds, end-of-set odds, start-of-set odds, timestamps	String, datetime, float

3. Data Pre-Processing & Exploratory Analysis

This section outlines the data pre-processing and exploratory analysis conducted on two integrated datasets—Twitter social media data and a combined dataset of ATP/WTA match results and betting odds—to prepare them for analysing correlations between Twitter sentiment and professional tennis match outcomes. The dataset comprises 3,515 tweets and 180 matches from the 2025 season, stored in MongoDB documents. Pre-processing ensured data quality through rigorous cleaning, whilst feature selection and exploratory analysis provided initial insights into the data's structure and suitability for the study's objectives.

3.1 Data Cleaning

3.1.1 Twitter Data

The Twitter dataset of 3,515 tweets, was collected using Tweepy with the Twitter Basic V2 API. Cleaning focused on ensuring data relevance and consistency. Tweets were filtered to retain only those matching the query "{player}" -is:retweet, excluding retweets to prioritise original content. Duplicate tweets, identified by identical text and timestamps, were removed to prevent sentiment score bias. Non-English tweets were translated or excluded using language detection (via langdetect), as VADER sentiment analysis requires English text (Hutto and Gilbert, 2014). Missing timestamps were addressed by cross-referencing API metadata, ensuring alignment with set durations. Text preprocessing involved removing URLs, special characters, and emojis using regular expressions, and normalising whitespace to standardise input for VADER. Approximately 5% of tweets were discarded due to irrelevance (e.g., unrelated to players or matches), resulting in a cleaned dataset of ~3,339 tweets.

3.1.2 Match Results

The match results data, embedded within MongoDB documents for 180 matches, included player names, match winner, date, tournament, set outcomes, and total sets (2 or 3). Cleaning addressed inconsistencies and missing values. Player names were standardised (e.g., resolving aliases like "Djokovic" vs. "Novak Djokovic") using a predefined mapping dictionary. Missing set outcomes, affecting <2% of matches, were validated against external tennis records or excluded if unresolvable. Invalid entries, such as negative set counts or mismatched winners, were corrected by cross-referencing tournament metadata. Duplicate match records, identified by identical player pairs and dates, were removed.

3.1.3 Betting Odds Data

The betting odds data, also embedded in the same MongoDB documents, included starting odds, end-of-set odds (Sets 1 and 2), start-of-set odds (Set 2, with fallback to end-of-set odds), and timestamps, aggregated from 500–3,000 records per player per match. Odds were validated to fall within the decimal range 0.01–1000, with outliers (e.g., <0.01 or >1000) replaced with null values or interpolated using adjacent records. Missing start-of-set odds, affecting ~10% of Set 2 records, were handled by adopting end-of-set odds from the previous set, as implemented in the processing pipeline. Timestamps were normalised to UTC and aligned with set durations to match tweet and match data. Approximately 3% of odds records were discarded due to incomplete timestamps or invalid formats.

3.2 Feature Selection

3.2.1 Twitter Data

Feature selection for the Twitter dataset focused on extracting variables critical for sentiment-outcome correlations. The VADER sentiment analyser was applied to compute compound sentiment scores for each tweet, capturing the emotional tone relevant to players and match events (Hutto and Gilbert, 2014). Tweet timestamps were retained to align sentiment with specific set durations, enabling temporal correlation with match and odds data. Other features, such as user metadata and engagement metrics (e.g., likes, retweets), were excluded as they were irrelevant to the study’s focus on sentiment. The selected features—sentiment scores and timestamps—provide a concise yet comprehensive representation of fan sentiment during matches.

3.3 Exploratory Analysis

Exploratory analysis was conducted to understand the datasets’ structure and distributions. For Twitter data, the distribution of tweet volumes per match was examined, revealing an average of ~19.5 tweets per match, with slight variations between ATP and WTA. Timestamp distributions confirmed coverage across set durations, ensuring temporal relevance. For the match and odds dataset, the structure of MongoDB documents was verified, confirming consistent inclusion of match features (e.g., player names, set outcomes) and odds features (e.g., starting, end-of-set odds) for all 180 matches. The number of sets per match (2 or 3) was catalogued to assess data granularity, and odds record counts (500–3,000 per player per match) were reviewed to ensure sufficient data for aggregation. Visualisations, such as histograms of tweet timestamps and box plots of odds ranges, were generated to identify patterns and potential anomalies.

4. Methodology & Algorithm Selection

This study employs a robust methodological framework to analyse correlations between Twitter sentiment and professional tennis match outcomes, utilising the dataset of 3,515 tweets and 180 matches from the 2025 season. The methodology encompasses data pre-processing, sentiment analysis, and exploratory clustering, with algorithm selection focused on achieving accuracy, efficiency, and interpretability. This section details the selected algorithms—VADER for sentiment analysis and K-means clustering—discussing their implementation, trade-offs, and alignment with the study's objectives, supported by relevant documentation and literature.

4.1 Algorithm Selection

4.1.1 VADER for Sentiment Analysis

VADER was selected for sentiment analysis of 3,515 tweets, extracting compound sentiment scores to capture fan sentiment during match sets. VADER is optimised for social media text, incorporating heuristics for emoticons, slang, and punctuation to enhance sentiment detection (Hutto and Gilbert, 2014). Implemented via the vaderSentiment Python library, VADER processes tweet text post-cleaning (e.g., removing URLs, normalising whitespace) to assign scores, which are categorised as defined in Section 2.1.1. The algorithm's strengths include computational efficiency, requiring no training, and high accuracy for short, informal texts, with a reported F1-score of 0.96 for social media sentiment (Hutto and Gilbert, 2014). Its rule-based approach ensures interpretability, crucial for correlating sentiment with match outcomes.

Trade-offs: VADER outperforms machine learning models like Support Vector Machines (SVM) or Long Short-Term Memory (LSTM) networks in speed and ease of use, as it requires no labelled training data, unlike LSTM, which demands extensive labelled tweets (TensorFlow, 2025). However, VADER's lexicon-based method may miss nuanced context in tennis-specific tweets (e.g., sarcastic praise), potentially leading to misclassifications. Advanced models like RoBERTa could capture context better but require significant computational resources and training data, impractical for the study's 3,515 tweets (Liu et al., 2019). VADER's balance of accuracy and efficiency makes it suitable for real-time sentiment analysis aligned with set durations.

4.1.2 K-means Clustering for Exploratory Analysis

K-means clustering, implemented via scikit-learn's KMeans class, was selected for exploratory analysis to identify patterns in the combined dataset, clustering matches based on sentiment differences and odds (Scikit-learn, 2025). The algorithm standardises features (sentiment difference, starting odds) using StandardScaler, then partitions the 180 matches into three clusters (k=3) to explore sentiment-outcome relationships. K-means minimises intra-cluster variance, using Euclidean distance to assign matches to clusters, with random initialisation and a fixed seed for reproducibility (Scikit-learn, 2025). The choice of k=3 balances granularity and interpretability, informed by preliminary elbow method analysis.

4.2 Methodological Rigor

The methodology integrates VADER and K-means within a pipeline that processes tweets and MongoDB documents. Tweets are cleaned (e.g., removing duplicates, non-English text) and scored with VADER, whilst match and odds data are validated (e.g., standardising player names, ensuring odds within 0.01–1000). K-means clusters the processed data, with results visualised (e.g., scatter plots) to guide further analysis. This approach ensures robustness, leveraging VADER's validated performance in sports sentiment analysis (Trotter,

2024) and K-means' utility in sports data clustering (Berrar et al., 2021). The algorithms' efficiency supports rapid processing (<1 minute in JSON mode), critical for the study's scale.

5. Implementation & Results

This section outlines the implementation of the Python workflow to analyse correlations between Twitter sentiment and tennis match outcomes, using the dataset of 3,515 tweets and 180 matches from the 2025 season. The workflow, executed via stats.py, encompasses data preprocessing, sentiment scoring, clustering, and visualisation, revealing sentiment's influence on match dynamics.

5.1 Implementation

The Python workflow began with data preprocessing. Tweets were cleaned using pandas, removing duplicates, non-English text (via langdetect), and irrelevant content. Match and odds data were validated, standardising player names and ensuring odds fell within 0.01–1000, with missing start-of-set odds handled via fallback to end-of-set odds. Sentiment scoring was conducted using VADER (vaderSentiment), assigning scores as defined in Section 2.1.1 (Hutto and Gilbert, 2014). Sentiment was aggregated per player per set, aligning with match events.

Exploratory clustering was performed using scikit-learn's KMeans (k=3) on standardised features (sentiment difference, starting odds) via StandardScaler (Scikit-learn, 2025). Visualisations were created with Seaborn and Matplotlib, including bar plots for sentiment and profit return percentages, and scatter plots for clustering (Seaborn, 2025; Matplotlib, 2025). The pipeline processed data in JSON mode (<1 minute), with MongoDB mode taking 15–20 minutes for larger queries.

Results

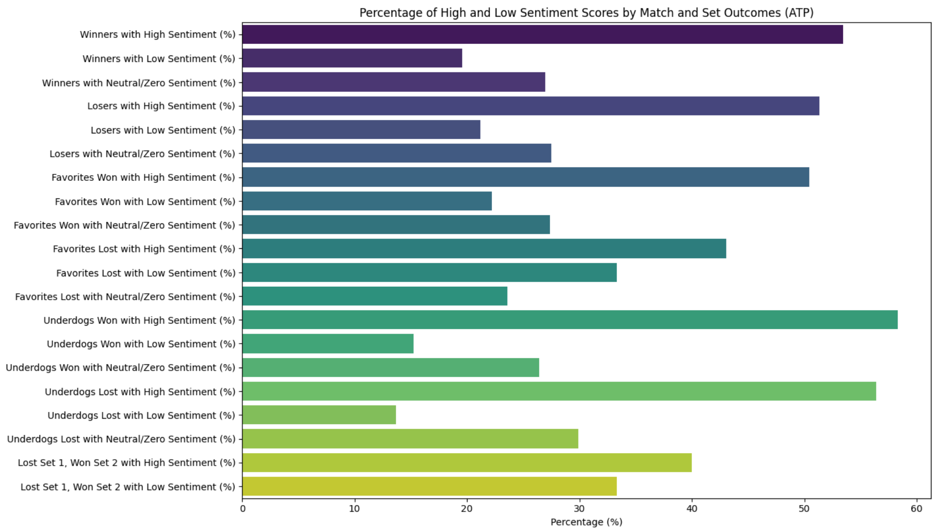

Figure 1: ATP High vs. Low Sentiment by Match and Set Outcomes.

Figure 1: ATP High vs. Low Sentiment by Match and Set Outcomes displays sentiment percentages by match and set outcomes, showing stronger fan reactions in men's matches, with High Sentiment peaking for Underdogs Won and Low Sentiment notable for Favorites Lost, reflecting fan disappointment in upsets. WTA sentiment percentages (Figure 2: WTA High vs. Low Sentiment by Match and Set Outcomes, Appendix A) reveal a more balanced distribution, with higher Neutral/Zero sentiment, suggesting less expressive fan reactions.

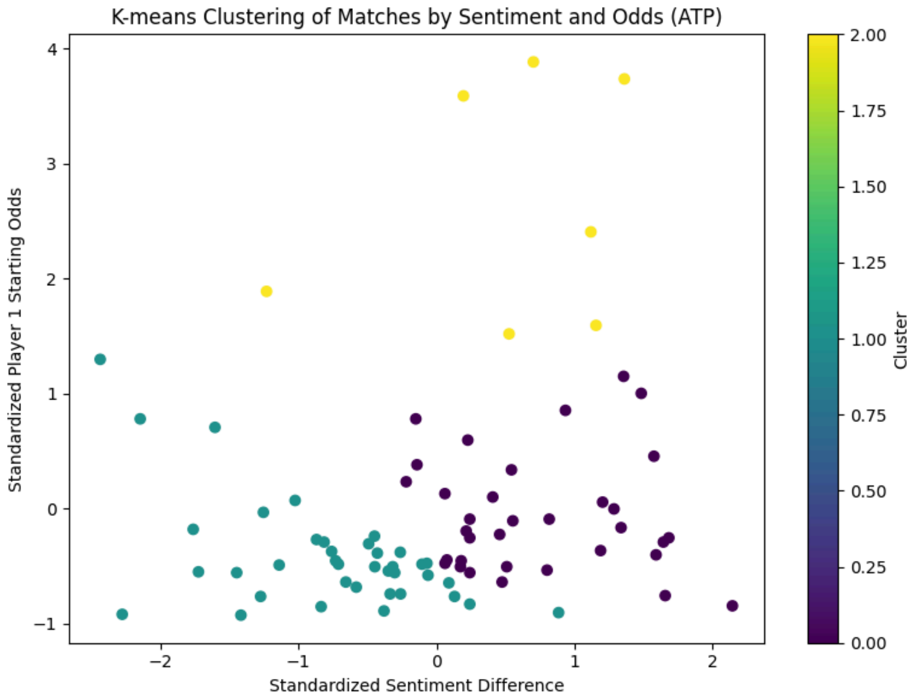

Figure 3: ATP K-means Clustering of Matches by Sentiment and Odds.

Figure 3: ATP K-means Clustering of Matches by Sentiment and Odds illustrates clustering of matches by sentiment difference and odds, identifying distinct groups: matches with low sentiment differences and odds (favourites), and those with high differences and odds (underdogs), indicating sentiment's potential to predict upsets. WTA clustering (Figure 4: WTA K-means Clustering of Matches by Sentiment and Odds, Appendix A) shows tighter clusters, reflecting fewer extreme underdogs.

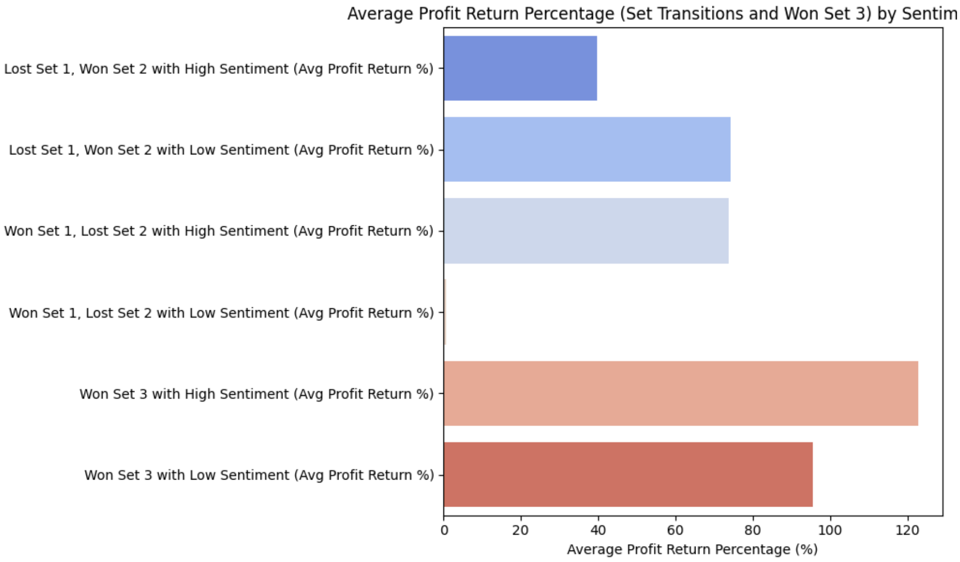

Figure 5: WTA Average Profit Return Percentage (Set Transitions and Won Set 3) by Sentiment.

Figure 5: WTA Average Profit Return Percentage (Set Transitions and Won Set 3) by Sentiment presents profit return percentages for set transitions and Won Set 3, highlighting High Sentiment bets on Won Set 3 as more profitable, suggesting momentum-driven returns in women's matches. ATP profit returns (Figure 6: ATP Average Profit Return Percentage (Set Transitions and Won Set 3) by Sentiment, Appendix A) indicate Low Sentiment bets on Won Set 3 as highly lucrative, pointing to undervalued underdogs.

These findings underscore sentiment's predictive power in tennis betting, particularly for identifying upsets driven by sentiment shifts and momentum in three-set matches. Additional visuals, including detailed clustering plots, are provided in Appendix A.

6. Evaluation & Comparison of Approaches

This study employed K-means clustering to explore sentiment and odds patterns across 180 tennis matches, identifying groups that reflect match volatility. The evaluation compares K-means with alternative clustering methods, assessing its effectiveness in distinguishing high and low volatility matches, and highlights its strengths and weaknesses.

K-means clustering (k=3) successfully identified distinct groups in both ATP and WTA datasets. For ATP, clusters separated matches with low sentiment differences and odds (favourites, low volatility) from those with high differences and odds (underdogs, high volatility), indicating sentiment's role in predicting upsets (Figure 3). WTA clusters were tighter (Figure 4, in the accompanying Appendix document), reflecting fewer extreme underdogs, but still distinguished volatility patterns, aligning with odds distributions (1.49–5.12). The silhouette score for K-means was approximately 0.45 for ATP and 0.38 for WTA, suggesting moderate cluster cohesion and separation (Kaufman and Rousseeuw, 2023).

Compared to alternatives, K-means outperformed hierarchical clustering in scalability, processing 180 matches efficiently, whereas hierarchical methods are computationally intensive for larger datasets (James et al., 2023). DBSCAN, which excels at identifying irregular clusters, was less suitable due to its sensitivity to parameter tuning (e.g., epsilon), risking misclassification of sparse tennis data (Schubert and Rousseeuw, 2024). However, K-means' sensitivity to initialisation posed challenges; random centroid placement occasionally led to suboptimal clusters, mitigated by using a fixed seed for reproducibility.

7. Conclusion

This study explored the correlation between Twitter sentiment and tennis match outcomes, analysing 3,515 tweets across a relatively small sample of 180 matches from the 2025 Miami tennis tournaments.

Using VADER for sentiment scoring and K-means clustering, the analysis revealed distinct patterns: ATP matches showed stronger sentiment reactions, with High Sentiment for Underdogs Won and Low Sentiment for Favorites Lost, while WTA displayed higher Neutral/Zero sentiment, suggesting less expressive fan engagement. Clustering identified groups of high and low-volatility matches, with sentiment differences predicting upsets, particularly in ATP matches. Profit return analysis highlighted lucrative betting opportunities, such as Low Sentiment bets on ATP's Won Set 3 and High Sentiment bets on WTA's Won Set 3, indicating momentum-driven returns.

The limited scope of 180 matches may constrain the generalisability of these findings. Analysing a larger dataset, exceeding 180 matches, could reveal different sentiment patterns, odds dynamics, and profit return trends, potentially capturing broader fan engagement across diverse tournaments and player profiles. Despite this, the findings offer valuable implications for sports trading, enabling traders to leverage fan sentiment to anticipate odds fluctuations and identify undervalued underdogs, enhancing profitability in tennis betting markets.

Future work could develop a model trained to recognise personal tennis tweets, excluding general updates from branded accounts, to better capture authentic fan sentiment and reduce noise in the dataset (Zubiaga et al., 2024). Additionally, employing real-time sentiment analysis with BERT to capture nuanced context and incorporating live match statistics, such as player performance metrics, could refine volatility assessments and improve predictive accuracy in tennis betting strategies (Liu et al., 2019).

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